
WorkBuddy

CODESYS Development

Skill Capability & Installation Guide

Skills required to replicate CODESYS PLC programming capability on a new WorkBuddy installation

Document Type	
	New WorkBuddy installation (any OS)
Skill Count	3 Skills (1 custom + 2 built-in)
	Easy - copy folder + install from marketplace

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1. Overview: What Skills Enable CODESYS Development

When you ask WorkBuddy to write CODESYS PLC code, the capability comes from a combination of the base AI model and installed Skills. To fully replicate this capability on a new WorkBuddy installation, the following 3 Skills need to be in place:

#	Skill Name	Type	Source	Purpose
1	codesys-development	Custom	Manual copy	CODESYS ST programming knowledge, FB design patterns, cam table algorithms
2	pdf	Marketplace	Built-in market	PDF document generation (reportlab), used for outputting guides and reports
3	skill-creator	Built-in	Built-in	Create new custom skills, needed if user wants to extend or modify skills

Capability Flow

The diagram below shows how these skills work together when you ask WorkBuddy to write CODESYS code:

Step	Skill Used	Action
1	codesys-development	Load IEC 61131-3 ST syntax rules, FB design patterns, cam table algorithms
2	(base AI model)	Generate ST code based on user requirements and loaded knowledge
3	(built-in Write/Edit)	Write code files to disk (.txt or .st files)
4	pdf	Generate PDF documentation (installation guides, reports)
5	skill-creator	(Optional) Create new skills if user wants to extend capabilities

2. Skill #1: codesys-development (Custom Skill)

This is the core custom skill that provides CODESYS programming knowledge. It contains IEC 61131-3 ST syntax rules, function block design patterns, cam table algorithms, and reusable code templates.

2.1 Skill Structure

File	Description
SKILL.md	Main skill definition: triggers, knowledge areas, workflow
references/fb-templates.md	Ready-to-use FB code templates (acquisition, processing, test program)
references/cam-table-patterns.md	Cam table design patterns: acquisition, fitting, smoothing algorithms
references/st-code-snippets.md	Common ST utilities: edge detection, state machine, array ops, type conversion

2.2 Knowledge Areas Covered

Area	Content
IEC 61131-3 ST Syntax	VAR_INPUT/OUTPUT/IN_OUT, data types, arrays, edge detection, RETURN, comments
FB Design Patterns	Data acquisition FB, processing/fitting FB, state machine FB
Cam Table Algorithms	Angle-triggered capture, average slope, linear fit, moving average smoothing
Naming Conventions	b=BOOL, n=INT, r=REAL, arr=ARRAY, Last/Next/Actual prefixes
Error Handling	Error code constants, parameter validation, overflow protection
VAR_IN_OUT Patterns	Large array passing by reference, caller declaration and invocation
Bidirectional Detection	Auto-detect forward/reverse, direction-aware trigger condition

2.3 Installation Method

This is a user-level skill. Install it on the new machine by copying the skill folder:

Method A: Direct Folder Copy (Recommended)

1. Locate the skill folder on the source machine:
C:\Users\\.workbuddy\skills\codesys-development\
2. Copy the entire codesys-development folder (including all subdirectories)

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3. Paste it into the target machine's skill directory:
`C:\Users\<new_username>\.workbuddy\skills\codesys-development\`
 4. If the `.workbuddy\skills\` directory does not exist, create it first.
 5. Restart WorkBuddy (or start a new conversation). The skill will be auto-detected.

Method B: Zip and Transfer

1. On source machine, zip the `codesys-development` folder:
Right-click `codesys-development` -> Send to -> Compressed (zipped) folder
2. Transfer the zip file to the new machine (USB, network, cloud)
3. On target machine, extract to:
`C:\Users\<username>\.workbuddy\skills\`
4. Verify the structure:
`.workbuddy\skills\codesys-development\SKILL.md`
`.workbuddy\skills\codesys-development\references\*.md`
5. Restart WorkBuddy.

Method C: Manual Creation

If folder transfer is not possible, you can have WorkBuddy recreate the skill on the new machine.

Open a new conversation in WorkBuddy on the target machine and type:

"Please create a skill called 'codesys-development' in my user skills directory. It should cover IEC 61131-3 Structured Text programming for CODESYS, including FB design patterns, cam table data acquisition and curve fitting, edge-triggered control, bidirectional detection, VAR_IN_OUT array binding, and error handling. Include reference files for FB templates, cam table algorithms, and ST code snippets."

WorkBuddy will use the built-in skill-creator to generate the skill automatically.

3. Skill #2: pdf (Marketplace Built-in)

This skill provides PDF generation and manipulation capabilities using Python libraries (reportlab, pypdf, pdfplumber). It is used to generate installation guides, reports, and documentation as PDF files.

3.1 Capabilities

Capability	Library	Use Case
Create PDF	reportlab	Generate structured PDF documents with tables, styles, Chinese fonts
Merge PDFs	pypdf	Combine multiple PDF files into one
Split PDF	pypdf	Split a PDF into individual pages
Extract text	pdfplumber	Extract text and tables from existing PDFs
Password protect	pypdf	Add encryption to PDF files

3.2 Installation Method

This skill is part of WorkBuddy's built-in document-skills marketplace package. It should be available by default on new installations. If not:

Method A: Check if already installed

1. Open WorkBuddy
2. Type: /skills
3. Look for 'pdf' in the skill list
4. If present, no action needed

Method B: Install from marketplace

1. In WorkBuddy, type: "Install the pdf skill"
2. Or use the find-skills command: "find a skill for PDF generation"
3. WorkBuddy will search the marketplace and install it automatically

Method C: Manual marketplace install

1. Open WorkBuddy Settings -> Skills / Connectors
2. Browse the built-in marketplace (cb_teams_marketplace)
3. Find 'document-skills' package
4. Install 'pdf' skill
5. Restart WorkBuddy

3.3 Python Dependencies

The pdf skill requires the following Python packages. WorkBuddy will install them automatically when the skill is first used. If manual installation is needed:

```
# Using WorkBuddy's managed Python venv
C:\Users\\.workbuddy\binaries\python\envs\default\Scripts\pip.exe install reportlab pypdf
pdfplumber pillow

# Or create the venv first if it doesn't exist:
C:\Users\\.workbuddy\binaries\python\versions\3.13.12\python.exe -m venv
C:\Users\\.workbuddy\binaries\python\envs\default
```

4. Skill #3: skill-creator (Built-in)

This skill is built into WorkBuddy and enables creating, modifying, and managing custom skills. It is needed if the user wants to create new skills or modify existing ones (like extending the codesys-development skill with new patterns).

4.1 Capabilities

Capability	Description
Create new skill	Generate a new SKILL.md with proper frontmatter, triggers, and reference files
Modify existing skill	Update SKILL.md content, add/modify reference files
Skill structure	Understands the directory structure: SKILL.md + references/ + scripts/ + assets/
Frontmatter	Generates proper YAML frontmatter (name, description, triggers)

4.2 Installation Method

This skill is built into WorkBuddy and requires no separate installation.

It is available by default on all WorkBuddy installations.

To verify it is available:

1. Open WorkBuddy
2. Type: /skills
3. Look for 'skill-creator' in the list
4. It should show as 'built-in'

4.3 Usage Example

To create or extend the codesys-development skill on a new machine:

User: "Create a skill called codesys-development for CODESYS ST programming..."

WorkBuddy will:

1. Load the skill-creator skill
2. Create the directory structure under ~/.workbuddy/skills/
3. Generate SKILL.md with proper frontmatter
4. Create reference files as needed
5. Confirm the skill is ready to use

5. Installation Summary Checklist

Use this checklist to verify all skills are installed on the new WorkBuddy machine:

Done	Skill	Method	Target Path	Time
[]	codesys-development	Copy folder	~/.workbuddy/skills/codesys-development/	~5 min
[]	pdf	Marketplace or /skills	(auto-installed)	~2 min
[]	skill-creator	Built-in (no action)	(built-in)	0 min

Quick Install Script (Windows PowerShell)

Run this in PowerShell on the new machine to quickly check and prepare the skills directory:

```
# 1. Create skills directory if it doesn't exist $skillsDir = "$env:USERPROFILE\.workbuddy\skills" if (-not (Test-Path $skillsDir)) { New-Item -ItemType Directory -Path $skillsDir -Force Write-Host "Created: $skillsDir" } else { Write-Host "Exists: $skillsDir" } # 2. Check if codesys-development skill exists $codesysSkill = "$skillsDir\codesys-development" if (Test-Path $codesysSkill) { Write-Host "[OK] codesys-development skill found" } else { Write-Host "[MISSING] codesys-development skill not found" Write-Host " -> Copy the codesys-development folder here: $codesysSkill" } # 3. Check for pdf skill (marketplace) Write-Host "Check /skills command in WorkBuddy for pdf and skill-creator availability"
```

Cross-Platform Notes

OS	Skills Directory	Notes
Windows	C:\Users\<user>\.workbuddy\skills\	Use backslashes in paths
macOS / Linux	~/.workbuddy/skills/	Use forward slashes; same folder structure

6. Verification: Test Your Setup

After installing all skills, verify by running this test conversation in WorkBuddy:

Test 1: CODESYS Skill Loaded

Type in WorkBuddy:

"Help me write a CODESYS function block that captures encoder positions triggered by winding axis angle every 10 degrees, with configurable array length."

Expected: WorkBuddy should generate proper IEC 61131-3 ST code with:

- VAR_INPUT with bEnable, EncoderPosition, WindingAxisAngle, StepInterval
- VAR_IN_OUT for array binding
- Edge-triggered control logic
- Bidirectional detection
- Error handling with error codes

Test 2: PDF Generation

Type in WorkBuddy:

"Generate a PDF document with a summary of the function block you just created."

Expected: WorkBuddy should use the pdf skill to generate a PDF file with:

- Structured content with tables
- Proper formatting
- File saved to the workspace directory

Test 3: Skill Management

Type in WorkBuddy:

"List all my installed skills."

Expected: WorkBuddy should list:

- codesys-development (user skill)
- pdf (marketplace skill)
- skill-creator (built-in skill)
- Any other installed skills

Troubleshooting

Issue	Cause	Solution
WorkBuddy doesn't know CODESYS	codesys-development skill not loaded	Verify folder exists at ~/.workbuddy/skills/codesys-development/SKILL.md; restart WorkBuddy

Issue	Cause	Solution
PDF generation fails	reportlab not installed in Python venv	Run: <code>pip install reportlab</code> in the managed venv
Skill not auto-detected	SKILL.md frontmatter malformed	Verify SKILL.md starts with <code>---</code> and has name/description fields
Chinese characters show as boxes	Missing font registration	Ensure SimHei/SimSun fonts exist at <code>C:\Windows\Fonts\</code>

If all three tests pass, the new WorkBuddy installation has full CODESYS development capability and can generate PLC code, documentation, and custom skills just like the original machine.